

HR Analytics – Employee Attrition Analysis & Prediction

Problem Statement

Employee attrition is a major challenge for organizations because it leads to increased recruitment costs, loss of productivity, and disruption of business operations. Understanding the factors that contribute to employee turnover is critical for improving retention strategies and workforce planning.

The objective of this project is to analyze employee data to identify the key drivers of attrition and build predictive models that can estimate the likelihood of an employee leaving the organization. The analysis focuses on demographic, job-related, compensation, and performance factors that may influence employee retention.

Project Objectives

- Perform data cleaning and preprocessing.
- Conduct Exploratory Data Analysis (EDA) to uncover attrition patterns.
- Identify factors strongly associated with employee turnover.
- Build and evaluate machine learning models for attrition prediction.
- Create an interactive dashboard to communicate insights and recommendations.

Expected Outcomes

The project will provide actionable insights into employee attrition, highlight high-risk employee groups, and support HR teams in designing effective retention strategies. Predictive analytics can help organizations proactively address turnover risks and improve workforce stability.